

$$Q6. A_{n \times n}, \quad p(A) = A^2 + 5A + I = 0$$

$$I = -A(A + 5I)$$

$$AB = I \Rightarrow B = A^{-1}$$

So A is invertible

$$Q7. \text{ for } \text{Nul}(T - \lambda I) \subseteq \text{Nul}(T^{-1} - \lambda^{-1} I)$$

$$TV = \lambda V \Rightarrow V = T^{-1}(\lambda V)$$

$$V = \lambda T^{-1}(V)$$

$$\Rightarrow \lambda^{-1} V = T^{-1}(V)$$

$$\text{For } \text{Nul}(T^{-1} - \lambda^{-1} I) \subseteq \text{Nul}(T - \lambda I)$$

$$T^{-1}W = \lambda^{-1} W \Rightarrow W = T(\lambda^{-1} W)$$

$$W = \lambda^{-1} T(W)$$

$$\Rightarrow \lambda W = T(W)$$

$$Q8. \quad T: P_2 \rightarrow P_2$$

Let $\mathcal{B} = \{1, x, x^2\}$ is a basis of P_2

$$T(1) = 1$$

$$T(x) = 2x + 1$$

$$T(x^2) = (2x + 1)^2$$

$$[T]_{\mathcal{B}} = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 2 & 4 \\ 0 & 0 & 4 \end{bmatrix}$$

$$\text{eigenvalues} = 1, 2, 4$$

$$\text{Corresponding eigen vectors} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}$$

respectively.

Hence $[T]_{\mathcal{B}}$ can be diagonalized by $\begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{pmatrix}$ ^{respectively.}

Now let

$$\mathcal{B}' = \{1, 1+x, 1+2x+x^2\}$$

then $[T]_{\mathcal{B}'} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 4 \end{pmatrix}$

Q9. Since S, T have distinct eigenvalues, so S, T are diagonalizable.

So there exist basis $\{v_1, v_2, v_3\}$ & $\{w_1, w_2, w_3\}$ s.t.

$$Sv_i = \lambda_i v_i \quad \& \quad Tw_i = \lambda_i w_i$$

where $\lambda_1 = 2, \lambda_2 = 6, \lambda_3 = 7$

Define

$$P: \mathbb{C}^3 \rightarrow \mathbb{C}^3$$

$$v_k \mapsto w_k \quad k=1,2,3$$

Clearly P is invertible.

Now

$$\begin{aligned} P^{-1}TPv_k &= P^{-1}Tw_k \\ &= P^{-1}(\lambda_k w_k) \\ &= \lambda_k v_k \\ &= Sv_k \end{aligned}$$

$$\Rightarrow P^{-1}TP\left(\sum_{i=1}^3 \alpha_i v_i\right) = S\left(\sum_{i=1}^3 \alpha_i v_i\right)$$

$$\Rightarrow P^{-1}TP = S$$

Q10. Take

$$S(x) = \begin{pmatrix} 2 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 6 & 0 \\ 0 & 0 & 0 & 7 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix}, T(x) = \begin{pmatrix} 2 & 1 & 1 & 1 \\ 0 & 2 & 1 & 1 \\ 0 & 0 & 6 & 1 \\ 0 & 0 & 0 & 7 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix}$$

S & T have 2, 6, 7 as eigenvalues & no others

T is not diagonalizable as $\dim(\text{Null}(T-2I)) = 1$

Suppose $\exists P \in L(\mathbb{C}^4)$ s.t. $S = P^{-1}TP$

Then $\mathcal{B} = \{Pe_1, Pe_2, Pe_3, Pe_4\}$ is a basis of $\mathbb{C}^4$

&

$$\begin{aligned} T(Pe_1) &= PS(e_1) \\ &= P(2e_1) \\ &= 2Pe_1 \end{aligned}$$

Similarly

$$T(Pe_2) = 2Pe_2$$

$$T(Pe_3) = 6Pe_3$$

$$T(Pe_4) = 7Pe_4$$

$\Rightarrow [T]_{\mathcal{B}}$ is diagonal, which is a Contradiction

$\Rightarrow$ No such P exists.